

Lab 01 –

Introduction to Programming Fundamentals

Programming Exercise:

1. Write a program for converting degree centigrade to Fahrenheit.

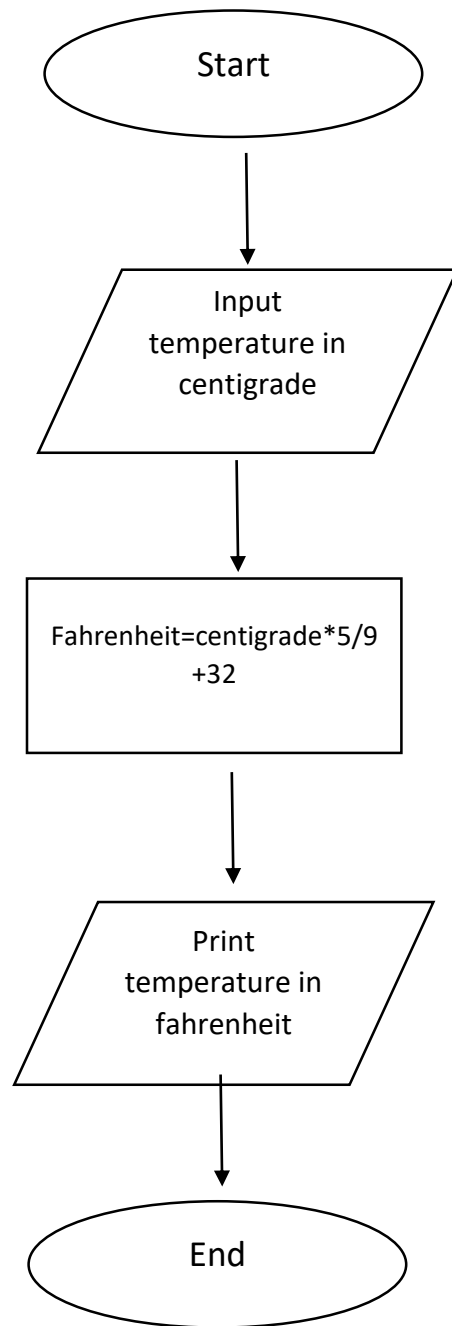
Source Code:

```
1  #PROGRAM FOR CONVERTING DEGREE CENTIGRADE TO FAHRENHEI
2  centigrade=float(input("Enter temperature in degree centigrade:"))
3  fahrenheit=centigrade*9/5+32
4  print("The temperature in degree fahrenheit is:",fahrenheit)
5
```

Output:

```
↑ C:\Users\hp\PycharmProjects\main\venv\Scripts\python.exe "C:\Users\hp\PycharmProjects\main\First Program.py"
↓ Enter temperature in degree centigrade:0
| The temperature in degree fahrenheit is: 32.0
|
| Process finished with exit code 0
```

Flowchart:



2. Write a program for converting degree Fahrenheit to centigrade.

Source Code:

 MAIN.py  First Program.py ×

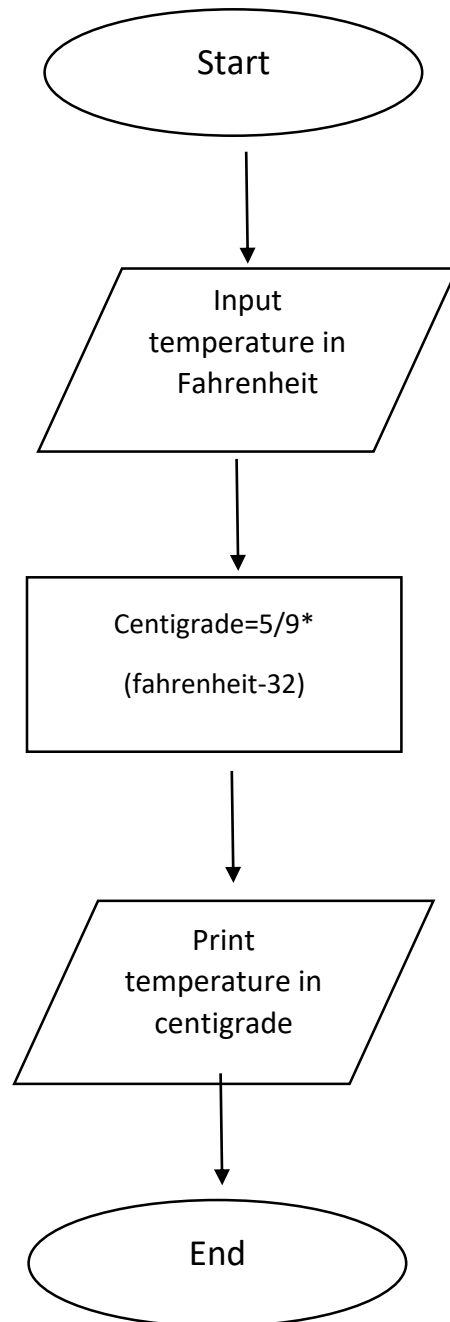
```
1  #PROGRAM FOR CONVERTING DEGREE FAHRENHEIT TO CENTIGRADE
2  fahrenheit=float(input("Enter temperature in degree fahrenheit:"))
3  centigrade=5/9*(fahrenheit-32)
4  print("The temperature in degree centigrade is:",centigrade)
```

Output:

 MAIN.py  First Program.py ×

```
1  #PROGRAM FOR CONVERTING DEGREE FAHRENHEIT TO CENTIGRADE
2  fahrenheit=float(input("Enter temperature in degree fahrenheit:"))
3  centigrade=5/9*(fahrenheit-32)
4  print("The temperature in degree centigrade is:",centigrade)
```

Flowchart:



3. Write a program to calculate area of the rectangle.

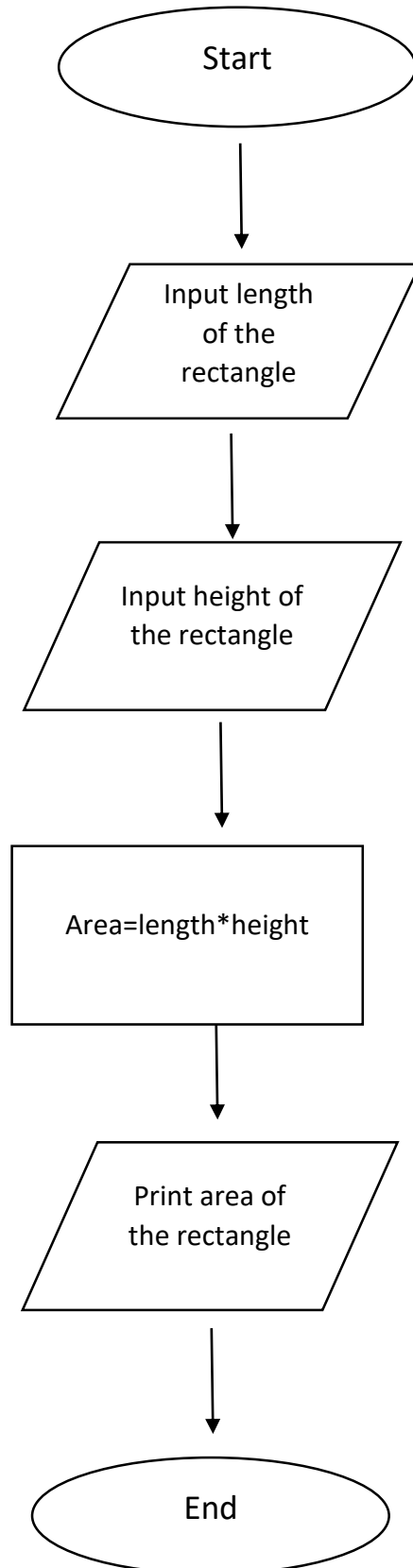
Source Code:

```
1  #PROGRAM TO CALCULATE AREA OF RECTANGLE
2  l=float(input("Enter length of the rectangle:"))
3  h=float(input("Enter height of the rectangle:"))
4  area=l*h
5  print("The area of the rectangle is:",area,"square units")
6
```

Output:

```
↑ C:\Users\hp\PycharmProjects\main\venv\Scripts\python.exe "C:\Users\hp\PycharmProjects\main\First Program.py"
↓ Enter length of the rectangle:3.7
Enter height of the rectangle:2.3
The area of the rectangle is: 8.51 square units
Process finished with exit code 0
```

Flowchart:

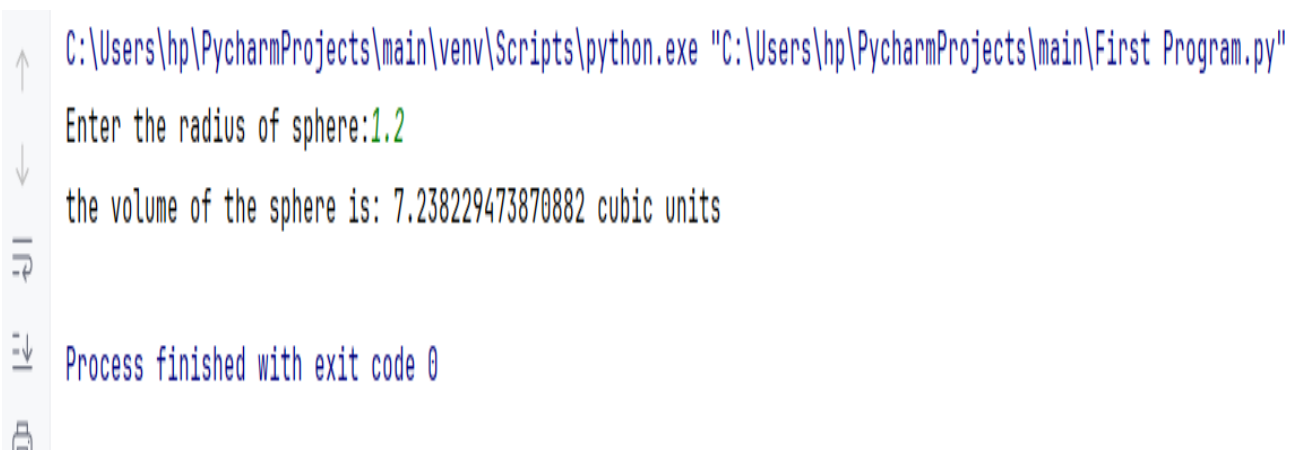


4. Write a program to calculate the volume of a sphere.

Source Code:

```
1  #PROGRAM TO CALCULATE THE VOLUME OF A SPHERE
2  import math
3  radius=float(input("Enter the radius of sphere:"))
4  volume=(4/3)*math.pi*radius**3
5  print("the volume of the sphere is:",volume,"cubic units")
,
```

Output:

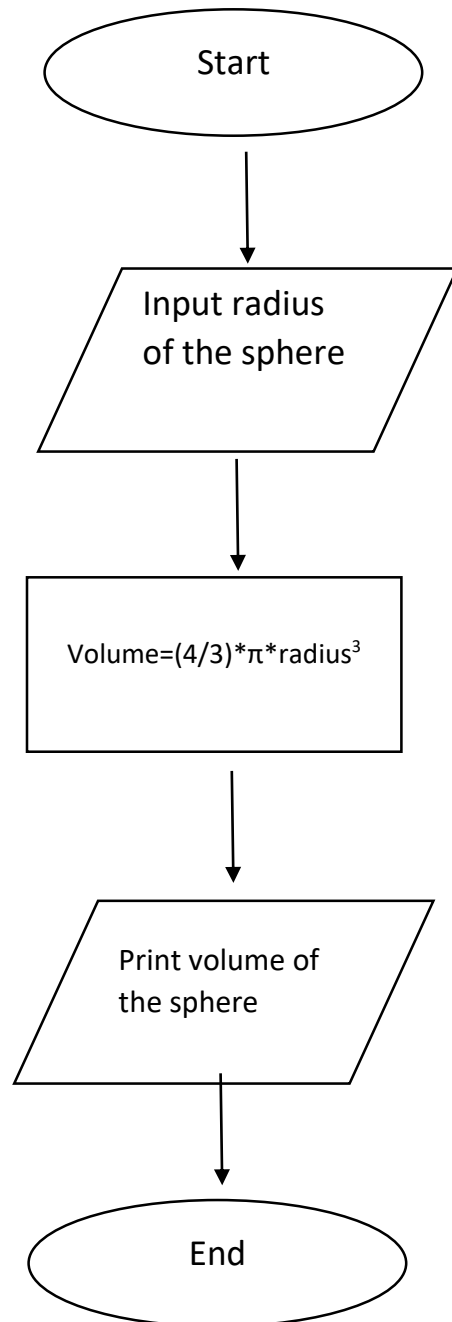


The screenshot shows a terminal window with the following content:

```
C:\Users\hp\PycharmProjects\main\venv\Scripts\python.exe "C:\Users\hp\PycharmProjects\main\First Program.py"
Enter the radius of sphere:1.2
the volume of the sphere is: 7.238229473870882 cubic units
Process finished with exit code 0
```

On the left side of the terminal window, there is a vertical toolbar with icons for navigating through the output history (up and down arrows, a search icon, and a list icon).

Flowchart:



5. Write a program that can write your name in upper case, lower case, and title case.

Source Code:

```
1  #program that can write your name in upper case, lower case, and title case.
2  name=input("Enter your name:")
3  #uppercase
4  uppercase_name=name.upper()
5  print("uppercase:",uppercase_name)
6  #lowercase
7  lowercase_name=name.lower()
8  print("lowercase:",lowercase_name)
9  #titlecase
10 titlecase_name=name.title()
11 print("titlecase:",titlecase_name)
```

Output:

```
C:\Users\hp\PycharmProjects\main\venv\Scripts\python.exe "C:\Users\hp\PycharmProjects\main\First Program.py"
```

```
Enter your name:raazia
```

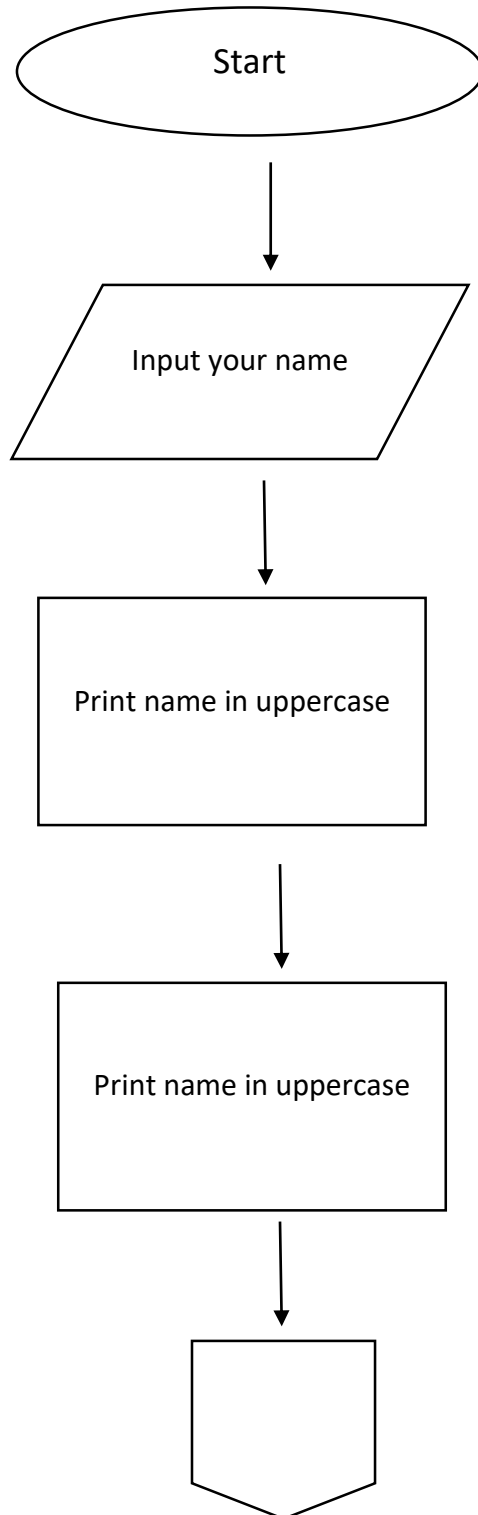
```
uppercase: RAAZIA
```

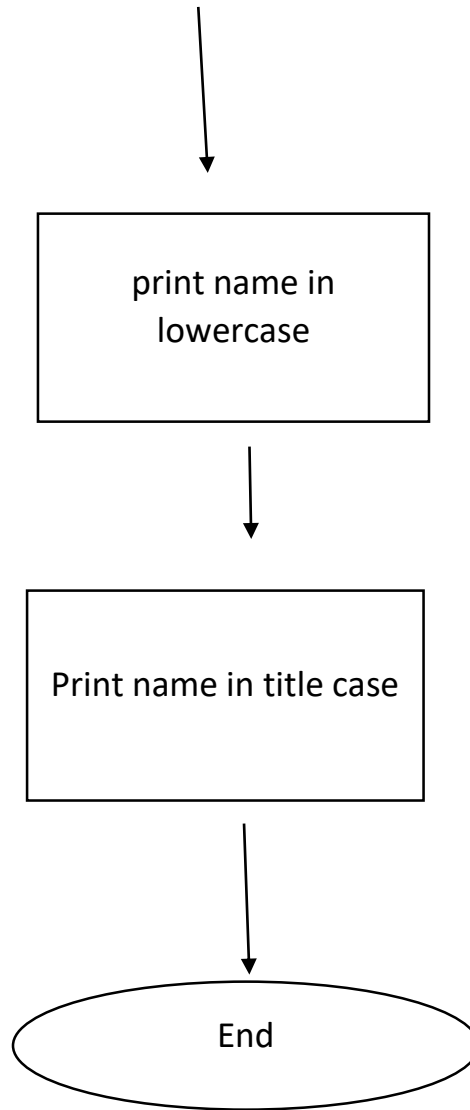
```
lowercase: raazia
```

```
titlecase: Raazia
```

```
Process finished with exit code 0
```

Flowchart:





6.the formula to calculate the compound interest annually is given by:

$$A=P(1+R/100)^t$$

$$\text{Compound Interest}=A-P$$

Where,

A is amount

P is the principal amount

R is the rate

T is the time span

Calculate the compound interest by taking input from the user by using above formula.

Source Code:

```
1  ✓ #PROGRAM TO CALCULATE COMPOUND INTEREST
2  #input values for principal amount, rate, and time
3  P=float(input("Enter the principal amount (P):"))
4  R=float(input("Enter the annual interest rate (R) in percentage:"))
5  t=float(input("Enter the time span (t) in years:"))
6  #calculate the amount after interest
7  A=P*(1+R/100)*t
8  #calculate compound interest
9  compound_interest=A-P
10 #display the result
11 print("The compound interest after", t , "years is:",compound_interest )
12 print("The amount after",t, "years is :",A )
```

Output:

```
C:\Users\hp\PycharmProjects\main\venv\Scripts\python.exe "C:\Users\hp\PycharmProjects\main\First Program.py"
Enter the principal amount (P):2000
Enter the annual interest rate (R) in percentage:2.5
Enter the time span (t) in years:2
The compound interest after 2.0 years is: 2100.0
The amount after 2.0 years is : 4100.0
Process finished with exit code 0
```

Flowchart:

